MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES, PLUG, PIN CONTACTS, DECADE INCREMENTS 10 THRU 70 CONTACT POSITIONS AND INTERMEDIATE POSITIONS OF 14, 24, 44, 54 AND 26, 36, 56, 66 FOR PRINTED WIRING BOARDS (.100 INCH SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.

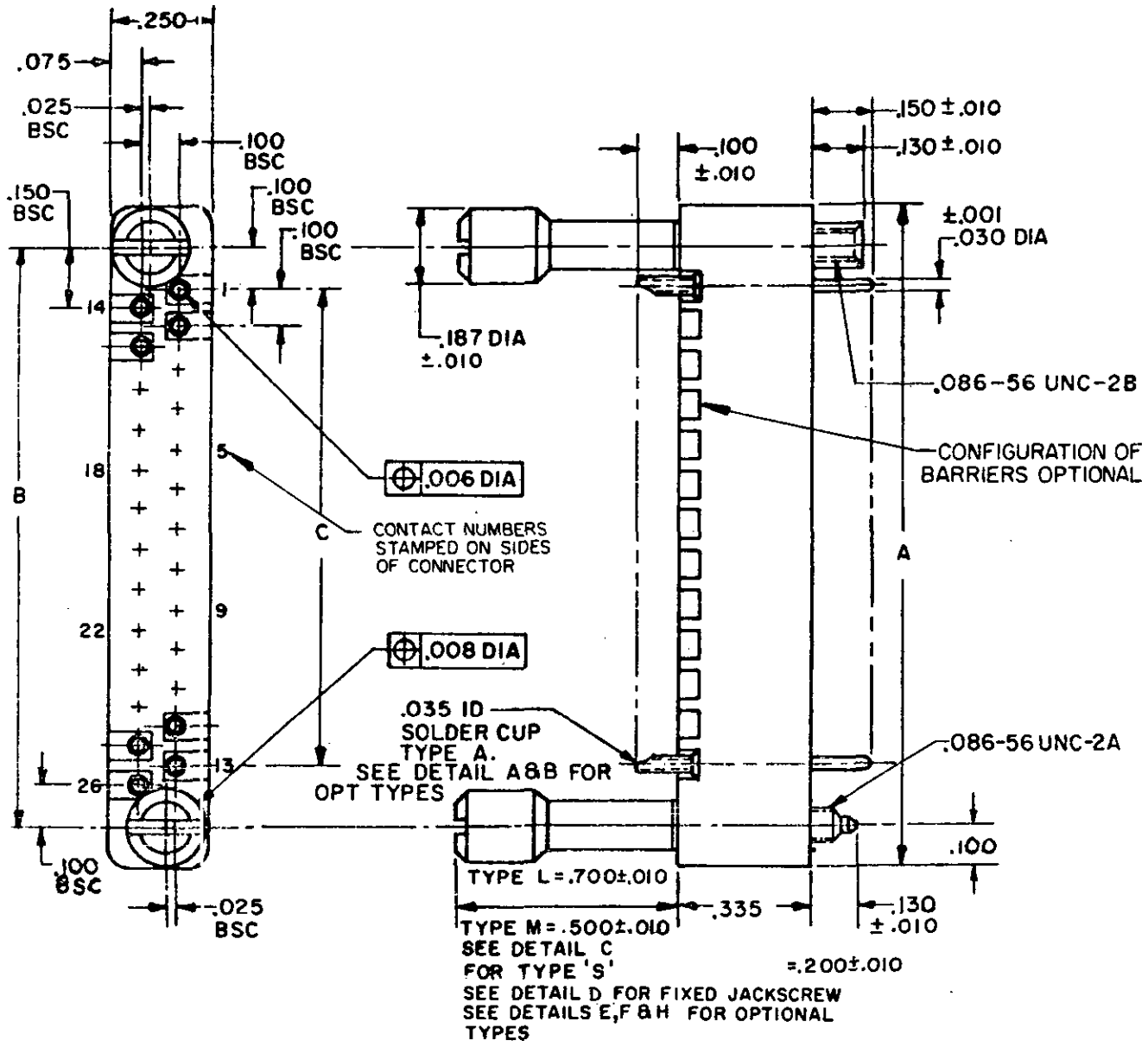
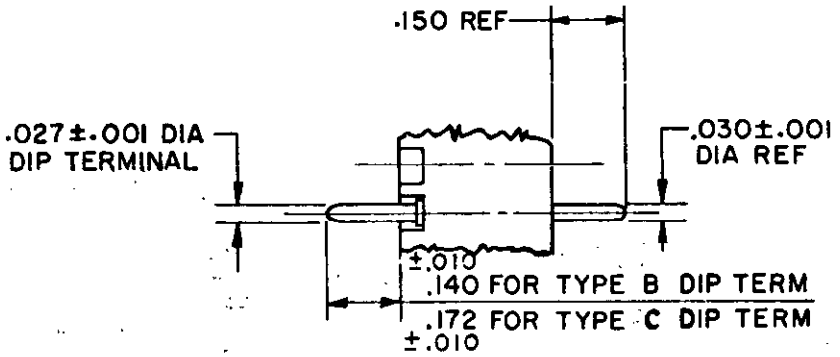


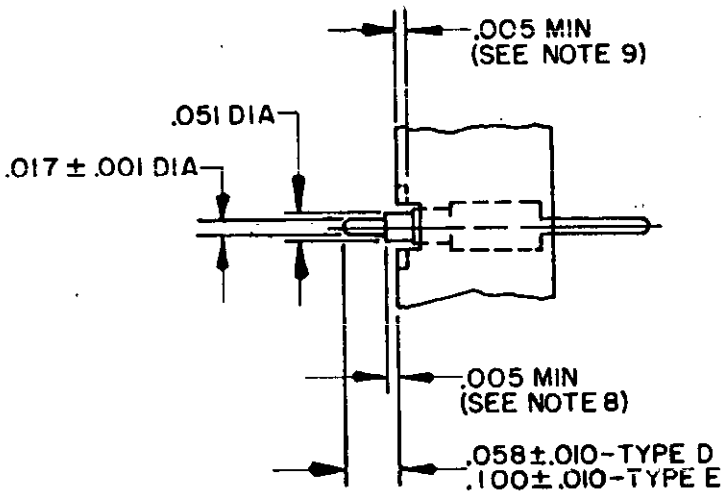
FIGURE 1. Connector, plug .100 (2.54 mm) spacing.

FSC 5935

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DETAIL A

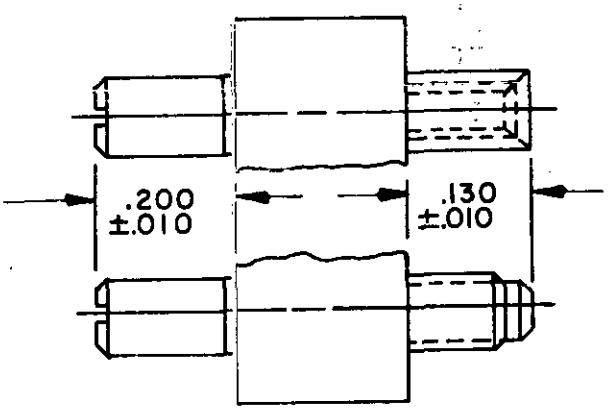


FLEXIBLE CIRCUIT TERMINAL

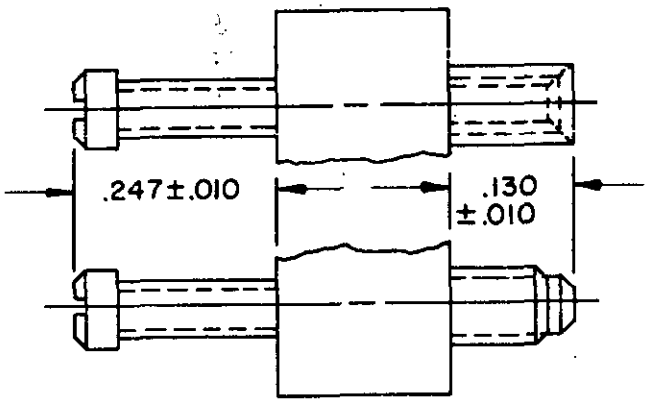
DETAIL B

FIGURE 1. Connector, plug .100 (2.54 mm) spacing - Continued.

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TYPE S SHORT TURNING
SLOTTED JACKSCREWS
DETAIL C

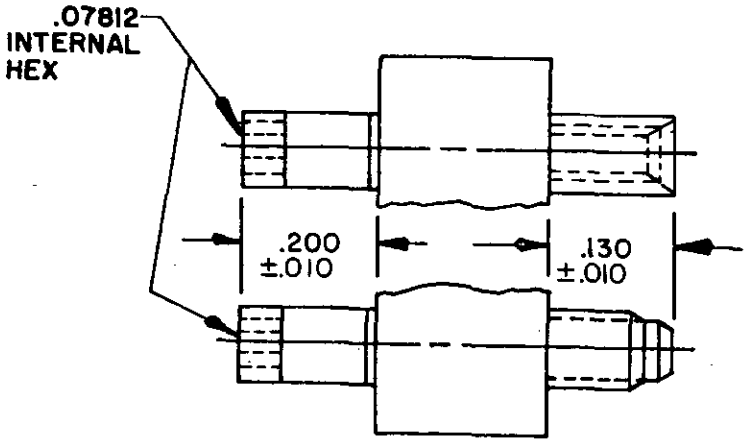


TYPE F FIXED JACKSCREWS

DETAIL D

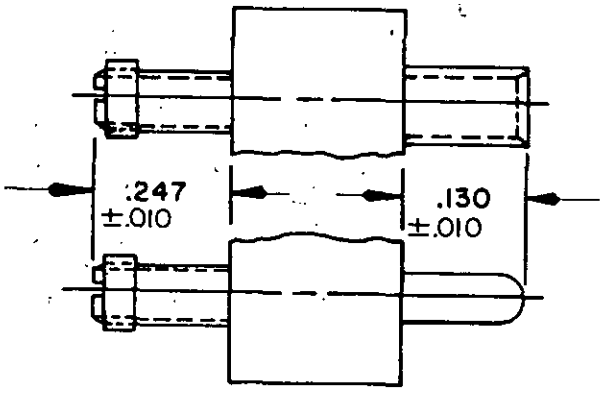
FIGURE 1. Connector, plug .100 (2.54 mm) spacing - Continued.

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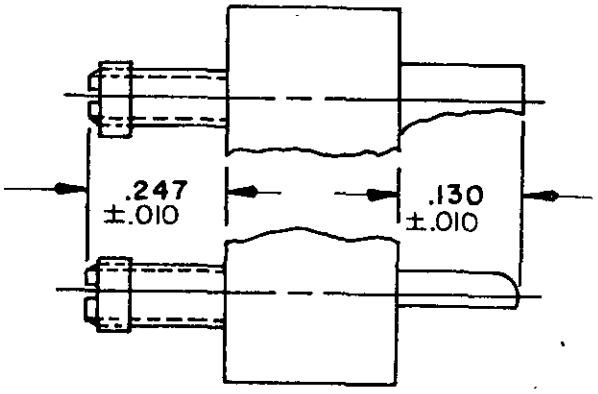
TYPE H SHORT TURNING
HEX JACK SCREWS

DETAIL E



TYPE X FULL ROUND GUIDE PIN
AND SOCKET

DETAIL F



TYPE Y D SHAPED GUIDE PIN
AND SOCKET

DETAIL H

FIGURE 1. Connector, plug .100 (2.54 mm) spacing - Continued.

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Number of contacts	Dimensions		
	A ±.010	BSC B	REF C
10	.850 (21.59)	.650 (16.51)	.400 (10.16)
14	1.050 (26.67)	.850 (21.59)	.600 (15.24)
20	1.350 (34.29)	1.150 (29.21)	.900 (22.86)
24	1.550 (39.37)	1.350 (34.29)	1.100 (27.94)
26	1.650 (41.91)	1.450 (36.83)	1.200 (30.48)
30	1.850 (46.99)	1.650 (41.91)	1.400 (35.56)
36	2.150 (54.61)	1.950 (49.53)	1.700 (43.18)
40	2.350 (59.69)	2.150 (54.61)	1.900 (48.26)
44	2.550 (64.77)	2.350 (59.69)	2.100 (53.34)
50	2.850 (72.39)	2.650 (67.31)	2.400 (60.96)
54	3.050 (77.47)	2.850 (72.39)	2.600 (66.04)
56	3.150 (80.01)	2.950 (74.93)	2.700 (68.58)
60	3.350 (85.09)	3.150 (80.01)	2.900 (73.66)
66	3.650 (92.71)	3.450 (87.63)	3.200 (81.28)
70	3.850 (97.79)	3.650 (92.71)	3.400 (86.36)

INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.035	.89	.172	4.37
.005	.13	.058	1.47	.187	4.75
.006	.15	.075	1.91	.200	5.08
.008	.20	.08712	1.9842	.247	6.27
.010	.25	.086	2.18	.250	6.35
.017	.43	.100	2.54	.335	8.51
.025	.64	.130	3.30	.500	12.70
.027	.69	.140	3.56	.700	17.78
.030	.89	.150	3.81		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
4. These connectors mate with connectors specified in MIL-C-55302/58, /62, /64, /65, and /66 using appropriate hardware.
5. Numbers indicating every fourth cavity are stamped on side with the exception of number one contact shall be marked.
6. Termination layout on .025 (0.64 mm) modular grid.
7. Metric equivalents are in parentheses.
8. Conformal coating clearances shall be .005 (0.13 mm) minimum (not required for optional design).
9. Optional design conformal coating clearance.

FIGURE 1. Connector, plug .100 (2.54 mm) spacing - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Material: Guide pins, type X guide sockets, jackscrews and jacksockets shall be made of corrosion resisting stainless steel per ASTM-A582 passivated per QQ-P-35 or shall be corrosion resistant copper nickel alloy (61 ±2% copper, 25 ±2% nickel, 2.5% max. other, and the balance zinc) and shall have a brinell hardness of 145 to 175. Type Y guide sockets shall be beryllium copper per QQ-C-550, nickel plated per QQ-N-290, class 2, 50 to 150 microinches.

Plating: The contact plating shall be MIL-C-55302 or gold per MIL-G-45204, type II, grade C, class 1, over nickel plating per QQ-N-290, class 2, 50 to 150 microinches in the engaging area and gold per MIL-G-45204, type II, grade C, class 00, over nickel plating per QQ-N-290, class 02, 50 to 150 microinches in the termination area.

Contact identification: See figure 1.

Mating and unmating:

Standard insertion force contacts: The maximum mating force, in pounds, shall be the number of contacts multiplied by 0.56 and the withdrawal force, in pounds, shall be a minimum of 0.08 times the number of contacts and shall not exceed the measured insertion force.

Low insertion force contacts: The maximum mating force, in pounds, shall be the number of contacts multiplied by 0.25 and the withdrawal force, in pounds, shall be a minimum of 0.04 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: The average contact resistance of all contacts measured shall not exceed 0.010 ohm, and no individual contact pair shall have a resistance exceeding 0.020 ohm.

Dielectric withstanding voltage:

Sea level: 1,000 volts rms.
High altitude: 300 volts rms.

Current rating: 5.0 amperes maximum per contact.

Polarization: Type X, regular, guide pin and socket, full round (see details E and F). Type Y, adjustable "D" shaped guide pin and/or socket (see MIL-C-55302/57, detail B). All connectors are supplied in the -1 polarized position. When guides are removed and reinstalled in another position apply Loctite 83-CV or equivalent to the threads. Ordering connectors in other than the -1 polarized position may be done by the OEM's by placing the desired polarized position in brackets three (3) spaces after the part number on the purchase order. Example M55302/63-A10Y [-32]. Polarization pins and sockets shall be adjusted to any desired configuration by spanner wrench M55302/57-01. Ordering connectors in the 1 polarized position without Loctite 83-CV or equivalent applied may be done by placing an "-0" in brackets three (3) spaces after the part number on the purchase order.

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Example of part number:

M55302/63	L-	A	10	L
Basic number of specification sheet	Type of contacts	Type of terminal (see figure 1)	Number of contacts (see figure 1)	Type of mounting hardware

L = Low insertion force contacts
S = Standard insertion force contacts

A = solder cup
B = dip terminal .140 long
C = dip terminal .172 long
D = .058 flexible circuit terminal
E = .100 flexible circuit terminal

L = .700 slotted turning jackscrew
M = .500 slotted turning jackscrew
S = .200 slotted turning jackscrew
F = fixed jackscrew
H = .200 hex turning jackscrew
X = full round
Y = "D" shaped

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - AS, TD
Air Force - 11, 85, 99
DLA - ES

User activities:
Army - AT, AV, MI
Navy - MC, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - CR

(Project 5935-3411-9)

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